Prabhavit Gupta

Forward Deployed Engineer | Applied AI, TypeScript, Python, Edge & Cloudflare Systems

+39 351 610 7807 | prabhavitg@gmail.com | linkedin.com/in/prabhavit | github.com/prab-gupta | Venice, Italy (EU Permanent Resident | Full Remote Europe | Immediate Availability)

Executive Summary

- **Forward Deployed Engineering & End-to-End Delivery:** Proven track record executing the complete lifecycle (*analyze* → *design* → *implement* → *release* → *operate*). Skilled in translating ambiguous client requirements into high-impact, shippable software increments through rigorous systems thinking, architecture decision records (ADRs), and iterative stakeholder demos.
- **Robust Full-Stack & Distributed Edge Architectures:** Deep technical expertise across modern application stacks: TypeScript/Node.js, Python (FastAPI/AsyncIO), React/Next.js, PostgreSQL schema design, Redis caching, and edge computing (Cloudflare Workers, KV, Durable Objects). Hands-on experience engineering scalable REST/GraphQL APIs, resilient data pipelines, and Dockerized microservices.
- **AI-Native Tooling, Agentic Patterns & Continuous Evals:** Practical builder of production AI solutions utilizing LLM orchestration (LangGraph, MCP - Model Context Protocol), function/tool calling, RAG pipelines, and automated evaluation frameworks (RAGAS, prompt regression suites) to ensure low latency, zero hallucinations, and deterministic business logic.

Technical Skills

- **Languages & Frameworks:** TypeScript, JavaScript (Node.js, Bun), Python (FastAPI, AsyncIO, PyDantic), React, Next.js, Tailwind CSS, SQL, Bash.
- **Edge Runtimes & Cloud Infrastructure:** Cloudflare (Workers, Durable Objects, KV, R2), AWS (Lambda, S3, ECS), GCP, Docker, GitHub Actions CI/CD, Terraform (IaC).
- **Data Systems & Messaging:** PostgreSQL, ClickHouse, Redis (caching, pub/sub, queues), Vector Stores (Qdrant, pgvector), Schema Design & Query Optimization.
- **AI Engineering & Agentic Blueprints:** Generative AI APIs (Anthropic Claude 3.5 Sonnet, OpenAI GPT-4o), Model Context Protocol (MCP), Tool Calling, Agentic Workflows (LangGraph, ReAct), Hybrid Search (BM25 + Dense RAG), RAGAS Evals.
- **FDE Practices & Delivery:** Architecture Decision Records (ADRs), Rapid Prototyping, Technical Specs, Client Demos, Observability, Unit/Integration Testing (Jest, PyTest, Playwright).

Professional Experience

Forward Deployed Engineer & AI Solutions Architect • *Independent / Applied AI Consultancy*
2024 – Present • Venice, IT (Remote Europe)

- Partnered directly with enterprise clients to scope, architect, and deploy custom AI solutions and internal platforms, turning vague operational bottlenecks into high-velocity production systems.
- Authored technical specifications, architecture blueprints, and ADRs; paired with product and domain leads to maintain continuous alignment and ship production-ready increments weekly.
- Engineered high-performance edge microservices utilizing Cloudflare Workers, Durable Objects, and TypeScript, achieving sub-40ms response times for mission-critical client workflows.
- Integrated custom Model Context Protocol (MCP) servers and tool execution harnesses, enabling autonomous LLM agents to safely interact with client PostgreSQL databases, internal APIs, and file repositories.
- Designed automated evaluation pipelines using quantitative metrics (faithfulness, context relevance) and regression harnesses, ensuring robust output validation prior to deployment.
- Implemented complete CI/CD automation via GitHub Actions and Docker containers, decreasing onboarding time for client engineering teams.

Full-Stack Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Developed and maintained distributed web services and interactive internal dashboards using Node.js, TypeScript, Next.js, and PostgreSQL.
- Architected asynchronous event processing pipelines using Redis queues and Python background workers for automated document ingestion, extraction, and validation.
- Designed and optimized relational schemas and indexing strategies in PostgreSQL, eliminating query bottlenecks and scaling transaction volume.
- Created reusable component libraries and boilerplate starter templates adopted by across-team engineers, accelerating internal feature delivery by 35%.
- Led client discovery sessions and technical demos, establishing strong feedback loops that directly informed product roadmaps and release cycles.

Key Engineering Projects

- **Autonomous MCP Tool Orchestrator (TypeScript / Python):** Built a client-facing agent framework implementing the Model Context Protocol (MCP) to provide LLMs with stateful, auditable tool invocation over corporate data stores, with strict schema verification.
- **Edge-Deployed Semantic Retrieval Gateway (Cloudflare Workers):** Designed a serverless hybrid search engine on Cloudflare Workers and PostgreSQL pgvector, orchestrating sparse BM25 indexing with dense embeddings to deliver real-time search across multi-tenant enterprise data.

Education & Languages

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous foundation in formal logic, systems modeling, game theory, and quantitative analytical methods.
- **Languages:** English (C1 Professional, fluent written/spoken), Italian (Working proficiency).
- **Work Authorization:** EU Long-Term Resident (Permesso di Soggiorno a Tempo Indeterminato, Italy) – Unrestricted legal right to work across the EU without sponsorship; Italian P.IVA / B2B contract ready.